

Experiment 04: Interfacing HC-SR04 Ultrasonic Sensor with Arduino UNO

Non-Contact Distance Measurement — Program Source Code & Experimental Output

1. Arduino C++ Source Code

Listing 4.1: Microcontroller Firmware for Ultrasonic Distance Measurement.

```
1 /*
2  * Experiment 04: Ultrasonic Distance Measurement using HC-SR04 and Arduino UNO
3  * Hardware Pin Connections:
4  *   - Arduino Digital Pin 8 -> HC-SR04 TRIG (Trigger Output)
5  *   - Arduino Digital Pin 7 -> HC-SR04 ECHO (Echo Pulse Input)
6  *   - Arduino 5V & GND      -> HC-SR04 VCC & GND Power Supply
7  * Serial Baud Rate: 9600 bps
8  */
9 const int trigPin = 8;    // Pin D8 connected to HC-SR04 TRIG terminal
10 const int echoPin = 7;    // Pin D7 connected to HC-SR04 ECHO terminal
11 long duration;            // Holds raw Time-of-Flight (ToF) in microseconds
12 int distance;             // Holds calculated one-way distance in centimeters
13
14 void setup() {
15     pinMode(trigPin, OUTPUT); // Configure Trigger pin as digital output
16     pinMode(echoPin, INPUT);  // Configure Echo pin as digital input
17     Serial.begin(9600);       // Initialize UART Serial Interface at 9600 Baud
18 }
19
20 void loop() {
21     // Step 1: Ensure a clean LOW signal before generating trigger pulse
22     digitalWrite(trigPin, LOW);
23     delayMicroseconds(2);
24     // Step 2: Transmit 10 microsecond Active-HIGH Trigger Pulse
25     digitalWrite(trigPin, HIGH);
26     delayMicroseconds(10);
27     digitalWrite(trigPin, LOW);
28     // Step 3: Measure HIGH pulse duration on ECHO pin (Time-of-Flight in us)
29     duration = pulseIn(echoPin, HIGH);
30     // Step 4: Calculate one-way distance in centimeters (Speed of sound = 0.034 cm/us)
31     distance = duration * 0.034 / 2;
32     // Step 5: Format and stream telemetry data to Serial Monitor
33     Serial.print("distance : ");
34     Serial.println(distance);
35     // Step 6: Measurement cycle delay to prevent acoustic multipath echo interference
36     delay(100);
37 }
```

2. Experimental Output & Serial Monitor Telemetry

A. Ultrasonic Distance Measurement & Calibration Log:

Trial	Actual Range	Serial Output	Measurement State & Hardware Observation
1	5.0 cm	distance : 5	Exact Setpoint Match (100% Correlation)
2	5.0 cm	distance : 4	Stable Reading with Tolerable Noise ($\Delta = -1.0$ cm)
3	5.0 cm	distance : 5	Exact Setpoint Match (100% Correlation)
4	10.0 cm	distance : 10	Linear Scaling Validated (100% Correlation)

B. Serial Monitor Stream (9600 Baud):

```
distance : 5
distance : 4
distance : 5
distance : 10
```